Module 4COSC004W Computer Systems Fundamentals
Test 2

Instructions:

1. You have **1 hour** to complete this test
2. This is a closed **book test**
3. You only have **ONE** attempt at this test – do not try to take the test again.
4. **Questions are randomized**, so do not bother trying to copy or help any of your colleagues.
5. Do **NOT** close the browser once you have started the test.
6. Do **NOT** navigate to any other site in that tab once you have started the test.
7. Do **NOT** use the browser *Back* button during the test.
8. All the questions are in the browser window in front of you, so scroll up and down to navigate between the questions.
9. You are **NOT** allowed to have any notes, lecture slides, tutorial exercises, practice test or any other materials with you during the test.
10. You are **NOT** allowed to use any communication device during the test
11. You are **NOT** allowed to communicate with anybody other than the tutor during the test.
12. You are **NOT** allowed to have more than one window open on your computer – that will be a browser with a single tab for the Blackboard test. Blackboard will open any extra required tabs as required.
13. Calculators are not allowed. The multiplication tables below are supplied to help you do the calculations quicker
14. The tables on the next page provide details of the relevant items in the Disk Sector Zero and the Partition Sector Zero. I have shaded the items that you are not required to use.

Multiplication tables:

1048576 table	
1	1048576
2	2097152
3	3145728
4	4194304
5	5242880
6	6291456
7	7340032
8	8388608
9	9437184
10	10485760
11	11534336
12	12582912
13	13631488
14	14680064
15	15728640

65536 table	
1	65536
2	131072
3	196608
4	262144
5	327680
6	393216
7	458752
8	524288
9	589824
10	655360
11	720896
12	786432
13	851968
14	917504
15	983040

4096 table	
1	4096
2	8192
3	12288
4	16384
5	20480
6	24576
7	28672
8	32768
9	36864
10	40960
11	45056
12	49152
13	53248
14	57344
15	61440

256 table	
1	256
2	512
3	768
4	1024
5	1280
6	1536
7	1792
8	2048
9	2304
10	2560
11	2816
12	3072
13	3328
14	3584
15	3840

16 table	
1	16
2	32
3	48
4	64
5	80
6	96
7	112
8	128
9	144
10	160
11	176
12	192
13	208
14	224
15	240

Disk Sector Zero – Partition Table Entry Locations		
Byte Range:	**Description:**	**Essential:**
0 – 445	Boot code	No
446 – 461	Partition Table Entry No. 1	Yes
462 – 477	Partition Table Entry No. 2	Yes
478 – 493	Partition Table Entry No. 3	Yes
494 – 509	Partition Table Entry No. 4	Yes
510 -511	Signature Value 0xAA55	Yes

Disk Sector Zero – Partition Table Entry Details		
Byte Range:	**Description:**	**Essential:**
0 – 0	Bootable flag	No
1 – 3	Starting CHS Address	No
4 – 4	Partition Type	Yes
5 – 7	Ending CHS Address	No
8 – 11	Starting LBA Address	Yes
12 – 15	Size in Sectors	Yes

Type values for DOS Partitions	
Type:	**Description:**
0x00	Empty
0x01	FAT12
0x04	FAT 16, 16 - 32 MB
0x06	FAT 16, 32 MB – 2GB
0x07	NTFS
0x0b	FAT32
0x0c	FAT32
0x82	Solaris x86
0x83	Linux
0xa5	FreeBSD
0xa6	OpenBSD
0xa8	Mac OSX

FAT Partition – Details of the First 36 Bytes in the FAT Partition Boot Sector	
Byte Range:	**Description:**
0 – 2	Assembly instruction to jump to boot code needed to boot the operating system
3 – 10	OEM Name in ASCII
11 – 12	Bytes per sector. Allowed values are 512, 1024, 2048 and 4096
13 – 13	Sector per cluster (data unit). Allowed values are powers of 2
14 – 15	Size in sectors of the reserved area
16 – 16	Number of FATs. Typically 2 for redundancy
17 – 18	Maximum number of files in the root directory for FAT12 and FAT16. This is 0 for FAT32 and typically 512 for FAT16.
19 – 20	16–bit value of number of sectors in the file system. If the number of sectors is larger than can be represented in this 2–byte value, a 4– byte value exists later in the data structure and this value should be 0 in that case
21 – 21	Media type. For fixed disks, value is 0xF8, for removable value is 0xF0
22 – 23	16–bit size in sectors of each FAT for FAT12 and FAT16. For FAT32, this field is 0
24 – 25	Sectors per track of storage device
26 – 27	Number of heads in storage device
28 – 31	Number of sectors before the start of partition
32 - 35	32–bit value of number of sectors in file system. Either this value or the 16–bit value above must be 0